Module Summary Report

Overview

This project involved performing exploratory data analysis (EDA) on the Customer Feedback and Satisfaction dataset to identify key factors influencing customer satisfaction. The dataset was sourced from Kaggle and includes demographic, behavioral, and feedback-related variables. The goal was to derive meaningful insights using structured analysis and reproducible workflows.

Dataset link: https://www.kaggle.com/datasets/jahnavipaliwal/customer-feedback-and-satisfaction

Github link - https://github.com/pranav-upd/ai-programming-foundations-project

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Dataset Description

The dataset contains 38,444 rows and 11 columns representing

customer demographics, purchasing behavior, and satisfaction metrics.

Key variables include:

Age, Income, ProductQuality, ServiceQuality, PurchaseFrequency, and SatisfactionScore. Additionally, categorical variables such as Gender, Country, FeedbackScore, and LoyaltyLevel

provide segmentation insights. The dataset is synthetic, meaning it is artificially generated but structured to resemble real-world customer data. The primary focus of the analysis was on understanding how product and service quality influence customer satisfaction.

Workflow Description

Ingestion

The dataset was loaded using Pandas from a CSV file into a DataFrame for analysis.

Cleaning

The dataset required minimal cleaning due to its synthetic nature. Column names were standardized, and checks were performed for missing values, duplicates, and data type consistency.

Exploratory Analysis

Custom EDA functions were created to compute summary statistics, grouped insights, and correlations between variables.

Visualizations

Matplotlib and Seaborn were used to generate distribution plots, scatter plots, and bar charts to explore relationships between variables.

Summary

Key insights were derived from both statistical outputs and visual patterns, particularly focusing on satisfaction drivers.

Key Decisions and Assumptions

- **Minimal Cleaning Approach:**

  Since the dataset contained no missing values or inconsistencies, only validation and standardization were performed. This aligns with best practices of avoiding unnecessary transformations that may distort data (Pimentel et al., 2019).

- **Focus on Correlation Analysis:**
  Correlation analysis was prioritized to identify relationships between variables, particularly those influencing satisfaction. This is a common exploratory step in data science workflows.

- **Visualization Choices:**
  - Distribution plots were used to understand data spread
  - Scatter plots to explore relationships
  - Bar plots to compare categorical groups
    These choices follow visualization best practices for clarity and interpretability (Wilke, 2019).

- **Assumption of Synthetic Data Reliability:**
  The dataset was assumed to be internally consistent, though not fully representative of real-world complexity.

Results and Interpretation

- **Strong Relationships:**

  ProductQuality and ServiceQuality showed strong positive correlations (~0.55) with SatisfactionScore.

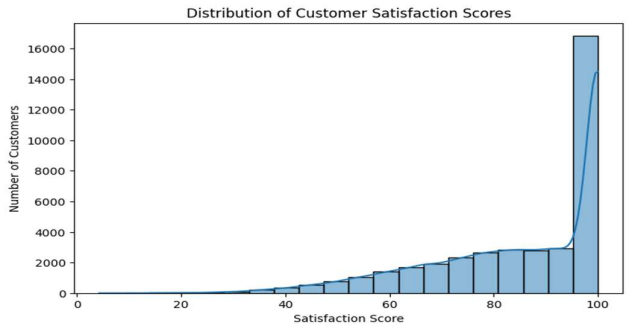

Fig 1

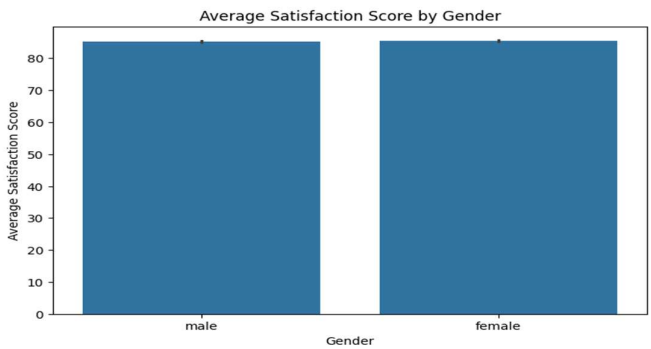

Fig 2

Figures 1 and 2 demonstrate these relationships.

- **Moderate Relationships:**

  Income and Age showed moderate correlations with satisfaction.

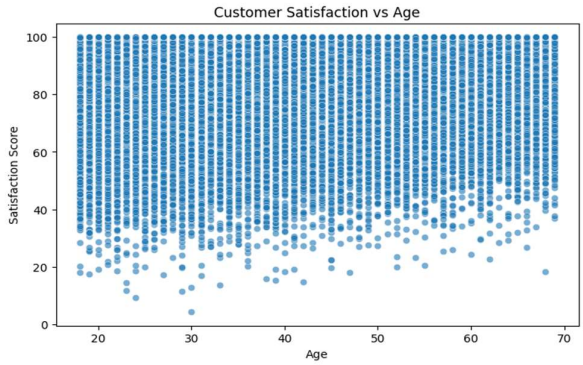

  Figure 3 highlights these trends.

- **Weak Relationships:**

  PurchaseFrequency had minimal correlation with satisfaction, indicating that frequent purchases do not necessarily imply higher satisfaction.

- **Key Insight:**

  Customer satisfaction is primarily driven by perceived quality rather than purchasing behavior.

Responsible Practice (Bias and Data Quality)

Although the dataset is clean, its synthetic nature introduces limitations. Synthetic datasets may not capture real-world biases, variability, or noise. Additionally, categorical simplifications (e.g., limited feedback levels) may reduce nuance in analysis.

Cleaning decisions can also introduce bias if data is removed or transformed improperly. In this project, minimal cleaning was performed to preserve original data characteristics. Future work could include validation against real-world datasets to improve generalizability.

Reproducibility

Reproducibility was ensured through:

- Use of a Jupyter Notebook (`data_workflow.ipynb`)
- Dependency management via `requirements.txt`
- Version control using Git (commits and branching)

These practices align with reproducible data science principles, ensuring that results can be consistently regenerated (Pimentel et al., 2019).

Sources and Citations

- Danchev, V. (2022). Reproducible Data Science with Python: An Open Learning Resource. Journal of Open Source Education, 5(56), 156. https://doi.org/10.21105/jose.00156

- Wilke, C. O. (2019). Fundamentals of Data Visualization. O'Reilly Media.